Apply filters to SQL queries

Project description

This project involves investigating potential security incidents and preparing employee machines for security updates by querying an organization's database using SQL. Through a series of tasks, I used filters such as `WHERE`, `LIKE`, `AND`, `OR`, and `NOT` to retrieve specific data from the `log_in_attempts` and `employees` tables. The goal is to identify suspicious login activity, filter login attempts by date, time, and location, and gather employee information for targeted security updates across various departments and offices.

Retrieve after hours failed login attempts

To identify failed login attempts that occurred after 18:00, I created a SQL query that filters the `log_in_attempts` table. The `WHERE` clause uses two conditions combined with `AND`: `login_time > '18:00'` ensures the attempts occurred after 6 PM, and `success = 0` (or `success = FALSE`) selects only failed attempts. This query helps investigate potential security incidents by focusing on unauthorized access attempts outside regular business hours.

SQL Query:

```
SELECT event_id, username, login_date, login_time, success
FROM log_in_attempts
WHERE login_time > '18:00' AND success = 0;
```

Retrieve login attempts on specific dates

To review login attempts on 2022-05-09 and 2022-05-08 due to a suspicious event, I used a SQL query with the `OR` operator. The `WHERE` clause filters the `log_in_attempts` table by checking if `login_date` equals `'2022-05-09'` or `'2022-05-08'`. This retrieves all login attempts (successful or failed) on these two dates, allowing a thorough investigation of the suspicious activity.

SQL Query:

```
SELECT event_id, username, login_date, login_time, success
FROM log_in_attempts
WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

Retrieve login attempts outside of Mexico

To investigate login attempts not originating from Mexico, I used the `NOT LIKE` operator with wildcards (`%`) to exclude records where the `country` column contains `'MEX'` or `'MEXICO'`. The `WHERE` clause combines two conditions with `AND`: `country NOT LIKE '%MEX%'` and `country NOT LIKE '%MEXICO%'`. This ensures all variations of Mexico are excluded, focusing the investigation on external suspicious activity.

SQL Query:

```
SELECT event_id, username, login_date, login_time, country, success
FROM log_in_attempts
WHERE country NOT LIKE '%MEX%' AND country NOT LIKE '%MEXICO%';
```

Retrieve employees in Marketing

To identify employees in the Marketing department located in the East building, I used `LIKE` with wildcards and the `AND` operator. The `WHERE` clause filters the `employees` table by checking `department = 'Marketing'` and `office LIKE 'East%'`, which matches any office value starting with `'East'` (e.g., East-170, East-320). This query provides a list of Marketing employees in the East building for targeted security updates.

SQL Query:

```
SELECT employee_id, device_id, username, department, office
FROM employees
WHERE department = 'Marketing' AND office LIKE 'East%';
```

Retrieve employees in Finance or Sales

To find employees in the Sales or Finance departments for a security update, I used the `OR` operator in the `WHERE` clause. The query filters the `employees` table by checking if `department = 'Sales'` or `department = 'Finance'`. This retrieves all relevant employee records, ensuring machines in these departments are updated.

SQL Query:

```
SELECT employee_id, device_id, username, department, office
FROM employees
WHERE department = 'Sales' OR department = 'Finance';
```

Retrieve all employees not in IT

To identify employees not in the Information Technology department for a security update, I used the `NOT` operator. The `WHERE` clause filters the `employees` table with `department NOT LIKE '%Information Technology%'`, excluding IT staff who already received the update. The use of `LIKE` with wildcards ensures variations of the department name are accounted for.

SQL Query:

```
SELECT employee_id, device_id, username, department, office
FROM employees
WHERE department NOT LIKE '%Information Technology%';
```

Summary

In this project, I applied SQL filters to investigate security incidents and support system updates. I retrieved after-hours failed login attempts, reviewed login activity on specific dates, and excluded attempts from Mexico to pinpoint suspicious events. Additionally, I identified employees in the Marketing, Sales, and Finance departments and excluded IT staff to prepare for targeted security updates. These tasks demonstrate my ability to use `AND`, `OR`, `NOT`, and `LIKE` in SQL to solve real-world security and administrative challenges.